

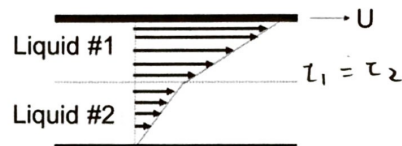
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ChE 311 Quiz 2 Spring 2019

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Closed book and closed notes
5 minutes, 10 points

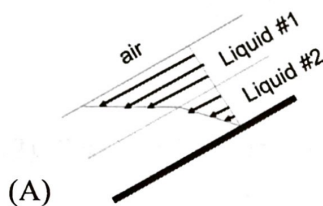
1. The velocity profile of the two immiscible liquids (#1 and #2) placed between two parallel plates, with top plate moving at constant velocity U and the bottom plate stationary, is shown in Figure 1.



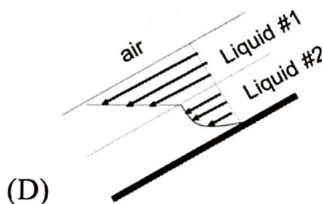
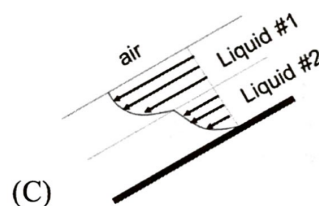
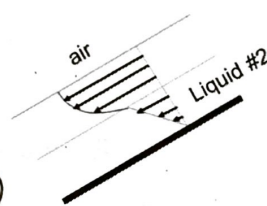
Now for both liquids flow down an inclined flat surface, which one of the following figures best describe the shear stress distribution and velocity distribution?

Answer: B

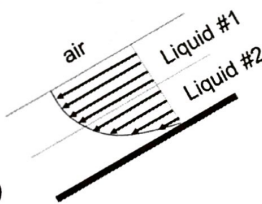
(C)



(B)



(E)



$\frac{1}{2} \tau$ $\int \rho g \sin \theta$

2. For Newtonian laminar flow in a straight circular tube, if the radius of the tube is reduced to be half and the volumetric flow rate keeps the same, how is the value of wall shear stress changed compared to the previous wall shear stress?

(E)

(A) $\frac{1}{2}$

(B) same

(C) 2

(D) 4

(E) 8

(F) 16

Answer: 8

3. Momentum fluxes.

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The combined momentum flux is the sum of the molecular momentum flux and the convective momentum flux,

$$\underline{\phi} = \rho \underline{\delta} + \underline{\tau} + \rho \underline{V} \underline{V}$$

An incompressible Newtonian liquid is flowing very slowly into a thin slot of thickness of $2B$ (in the y -direction) and with W (in the z direction). The velocity distribution within the slot is,

$$v_x = a \left[1 - \left(\frac{y}{B} \right)^2 \right]$$

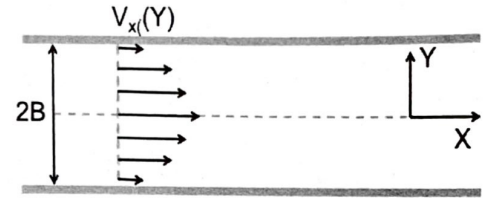
Apparently $v_y = v_z = 0$.

The pressure is a linear function of x , $P(x) = -cx$.

The viscosity of the fluid is μ . And a and c are the constant parameters.

(1) Please evaluate ϕ_{xx} at $x = 0$ and $y = B$. $\Rightarrow \tau_{yx} = 0$

$$\begin{aligned} \phi_{xx} &= \int v_x v_x + P \delta_{xx} + \tau_{xx} \\ &= \int a^2 \left(1 - \left(\frac{y}{B} \right)^2 \right)^2 - cx + \tau_{xx} \\ &= -cx = 0 \end{aligned}$$



$$\delta = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(2) Please evaluate ϕ_{xy} at $x = 0$ and $y = B$.

$$\begin{aligned} \phi_{xy} &= \int v_x v_y + P \delta_{xy} + \tau_{xy} \\ &= 0 \end{aligned}$$

(3) Please evaluate ϕ_{xx} at $x = 0$ and $y = 0$.

$$\begin{aligned} \phi_{xx} &= \int v_x v_x + P \delta_{xx} + \tau_{xx} \\ &= \int a^2 - cx + \mu \frac{a}{B} \end{aligned}$$

$$\tau_{yx} = -\mu \frac{\partial v_x}{\partial y}$$